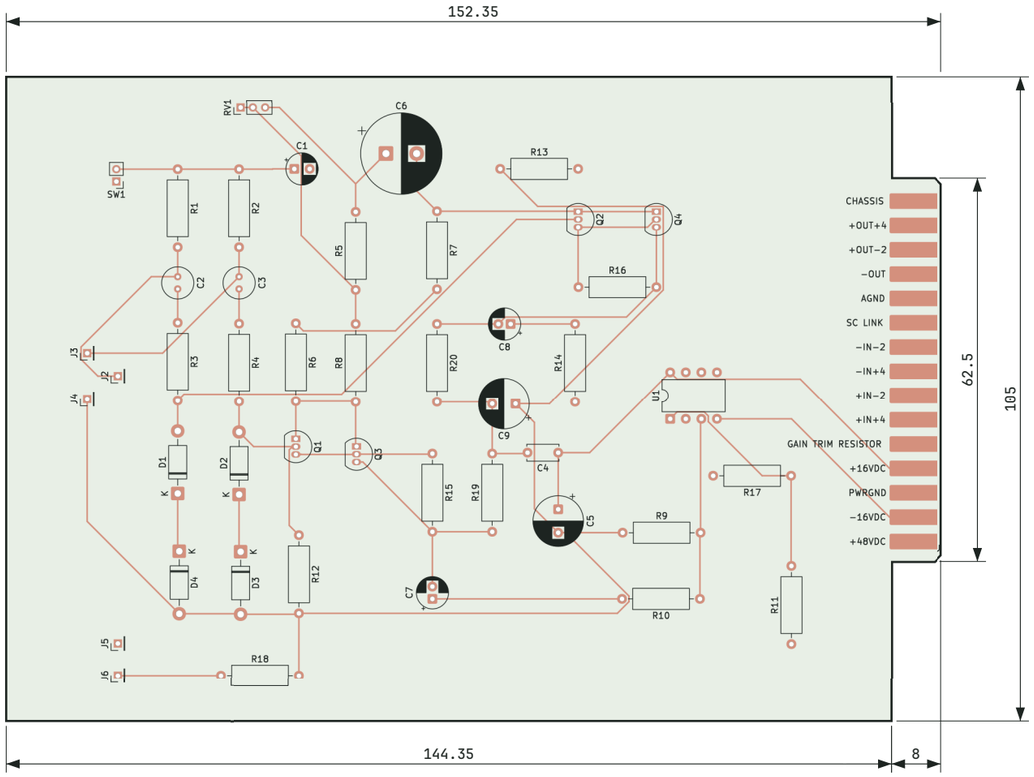
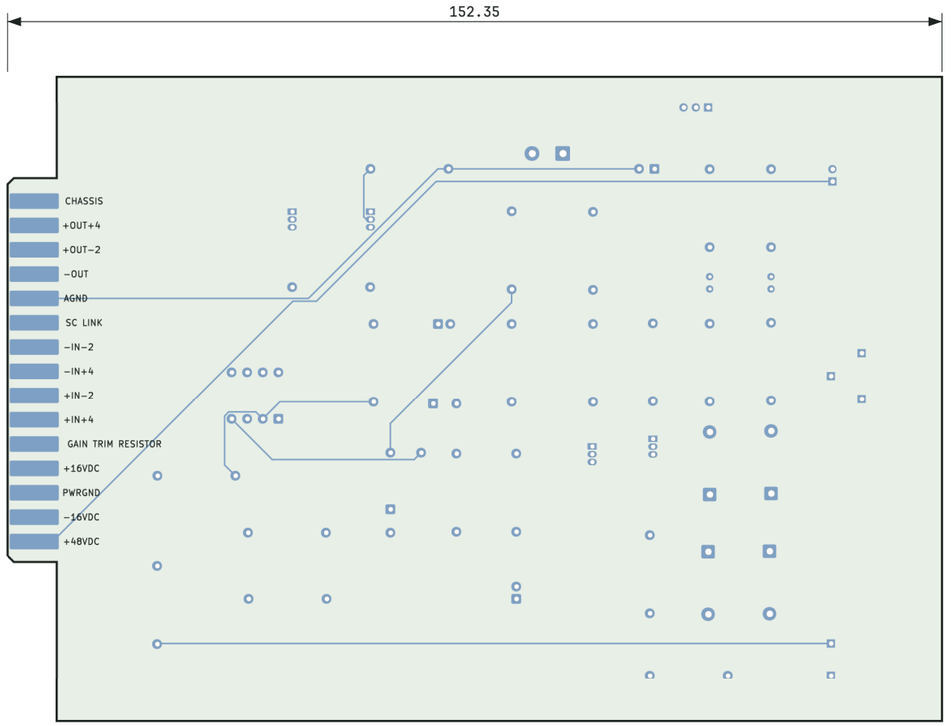
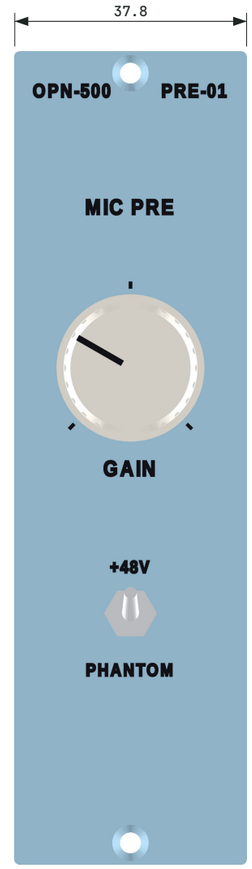


A TOP · COMPONENT SIDE SCALE 1:1

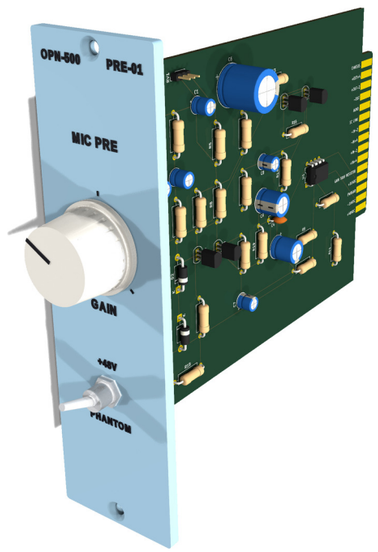


B BOTTOM · SOLDER SIDE (MIRRORED) SCALE 1:1



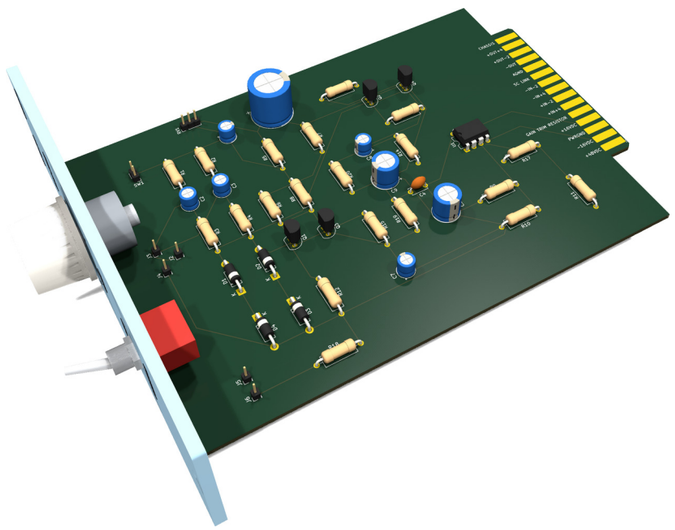
C MOCK-UP PANEL 1:1

for the demo only, not a design file



D PRE-AMP WITH MOCK-UP PANEL 3D

GAIN knob and +48V toggle



E PRE-AMP CARD 3D

component side

BOARD DATA

SIZE	152.35 × 105 mm
STACKUP	2 layers · FR-4 1.6 mm
PARTS	46 (45 through-hole, 1 SMD)
MIN TRACK	0.25 mm
MIN DRILL	0.6 mm
VIAS	0
MOUNTING	none on the card; held by the rack edge connector and panel
AUDIO	mic in and line out on J2-J6 flying leads

NOTES

1. Print at 100 % (actual size) on A3: views A and B are 1:1. Dimensions in mm.
2. Circuit after ESP Project 66 (mic preamp) with Project 96 phantom power; the schematic carries no values, and the build uses the ESP values.
3. Copper: top layer red, bottom layer blue, holes white. Routed on both layers with no copper pour.
4. The front panel (C, D) is a mock-up for the demonstration, styled as the Compressor's printed faceplate: the repo has no panel yet. GAIN (RV1) and the +48 V switch (SW1) wire to the card on flying leads.
5. Card edge tidied for this drawing: the fingers are labelled pin for pin. On the repo board the labels sit one pin towards pin 15, and the connector footprint carries a second edge outline 0.1 mm off the card's.

UTS 500
SERIES

UTS 500 SERIES · API 500 MODULE · PCB DRAWING

SHEET 2 / 3

PRE-AMP

Microphone preamplifier card · ESP P66 + P96 phantom power

SOURCE github.com/UTS-500-Series/Pre-Amp	REVISION 34efc53 · board 2026-09-18	DATE 2026-10-21
BOARD Series-500.kicad_pcb	SCALE 1:1 unless noted	SIZE · UNITS A3 · mm
LAYERS 2 copper	TOOL KiCad 10	LICENCE CERN-OHL-S v2